Anomaly Detection in Graphs via Topology-Aware Attention Mechanisms

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***Abstract*—Anomaly detection in graph-structured data is critical for a wide range of security-sensitive applications, including fraud detection, intrusion monitoring, and system fault analysis. While Graph Neural Networks (GNNs), particularly Graph Attention Networks (GATs), have demonstrated strong capabilities in modeling complex relational structures, they remain vulnerable to adversarial perturbations and often fail to capture global structural patterns, such as symmetries and higher-order connectivity. These limitations reduce their effectiveness in identifying subtle or distributed anomalies that arise beyond local neighborhoods.**

In this work, we propose a novel framework that enhances the robustness, interpretability, and anomaly detection capabilities of GATs by integrating topological features derived from persistent homology into the attention mechanism. By embedding persistent topological descriptors into the computation of attention scores, our model becomes more sensitive to structural irregularities that signal potential threats or inconsistencies particularly those emerging from global graph geometry. We evaluate our approach on both synthetic and real-world datasets, and demonstrate that our topologically-aware attention mechanism consistently outperforms traditional GNN baselines. These results underscore the value of combining topological data analysis with attention-based graph learning for secure, and structure-aware anomaly detection.

***Index Terms*—Anomaly Detection, Graph Neural Network (GNN), Attention Mechanism, Topological data analysis, Persistent homology.**

I. INTRODUCTION

Graph-structured data enables a wide range of critical applications, ranging from financial transaction networks and communication networks, through biological systems and social networks, to knowledge bases. Modeling these complex, interconnected data structures requires machine learning methods that can reason about both the content of individual entities (nodes) and the relationships between them (edges). Graph Neural Networks (GNNs) have emerged as a powerful paradigm for learning from such data, because they have the ability to iteratively aggregate information from the neighborhood of a node to produce expressive node, edge, or graph-level representations. Among the many varieties of GNNs, some prominent architectures have been developed to capture the structure and semantics of graphs through neighborhood aggregation. Graph Convolutional Networks (GCNs), for example, extend the convolution operation to non-Euclidean spaces by aggregating local information from an immediate neighbor of a node in terms of normalized adjacency matrices. GCNs are efficient and effective on semi-supervised node classification problems, but their aggregation mechanism is fixed and does not necessarily reflect node-level heterogeneity. GraphSAGE introduces a more flexible approach by learning aggregation functions (e.g., mean, LSTM, pooling) that generalize to new nodes in inductive settings. Conversely, Message Passing Neural Networks (MPNNs) unify many GNN variants by casting node updates as a two-step message passing and aggregation process that allows for a rich class of node and edge feature interactions that are learnable. Yet each of these methods typically relies on uniform or fixed aggregation schemes that prove not robust enough when processing graphs with very irregular or dynamic local structures.

To overcome these challenges, Graph Attention Networks (GATs) introduce a data-driven attention mechanism that allows each node to assign adaptive importance weights to its neighbors based on learnable attention scores. This selective aggregation enables GATs to dynamically prioritize relevant connections, improving both expressiveness and interpretability. GATs have shown strong performance across various domains, including node classification, recommendation systems, and knowledge graph completion, particularly in scenarios with noisy or heterogeneous graph structures.

However, despite their local adaptability, GATs share a fundamental limitation with other neighborhood-based GNNs: a lack of global structural awareness. Operating primarily on local neighborhoods, GATs may overlook higher-order patterns such as symmetry, global motifs, or topological redundancy structures that are often critical for understanding graph behavior in complex systems. For example, they may fail to distinguish between a regular graph and one perturbed by an adversary in a symmetric way. Moreover, although GATs produce attention scores, these are typically based on local feature similarity and offer limited insight into the graph's overall geometry.

This shortcoming is particularly pronounced in security-critical applications, where GNNs have been increasingly adopted for tasks such as fraud detection, intrusion detection,

malware propagation analysis, and user behavior modeling. In such domains, the goal is to detect anomalies, structural or behavioral deviations that may signal adversarial attacks, system failures, or policy violations. GNNs are well-suited to this task due to their capacity to fuse relational and feature-based information. In particular, GATs are appealing because their attention mechanism helps isolate suspicious local behaviors.

Nevertheless, existing GAT-based models tend to underperform when anomalies are subtle, distributed, or global. For instance, coordinated fraud schemes often involve multiple benign-looking nodes forming suspicious cycles or redundant subgraphs; stealthy intrusions may exploit regularities in communication patterns. Because GATs are primarily sensitive to local interactions, they often fail to detect such globally emergent behaviors.

To address this gap, we propose a novel framework that integrates Topological Data Analysis (TDA), specifically persistent homology, into the GAT architecture. Persistent homology provides a principled way to extract and summarize the multi-scale topological features of a graph, including connected components, loops, and voids. These features capture the shape of the graph and remain stable under small perturbations, making them ideal for identifying structural anomalies. For example, persistent cycles may reveal hidden feedback loops in networks, while topological changes in network communication graphs may indicate abnormal activity.

Our approach embeds persistent topological descriptors into the GAT attention computation, allowing the model to incorporate both local feature information and global structural context. This topologically-informed attention mechanism enables GATs to better detect anomalies that arise from structural irregularities, symmetric perturbations, or emergent global patterns without sacrificing their strength in modeling local interactions.

We evaluate our method on both synthetic benchmarks and real-world datasets, including Cora and Amazon datasets. Our results demonstrate that the topologically-aware GAT model outperforms standard baselines, including GCN, GAT, and TOGL, in anomaly detection tasks, particularly in identifying non-local anomalies. Additionally, our model improves interpretability by linking anomalous behavior to persistent topological features, providing geometric insights into the model's predictions.

In summary, our contributions are as follows:

- We identify fundamental limitations of attention-based GNNs in detecting structural and symmetric anomalies due to their local focus
- We propose a novel method that incorporates persistent homology-derived features into the GAT attention mechanism to improve global structural awareness.
- We empirically demonstrate that our method improves anomaly detection performance across synthetic and real-world datasets.
- We offer a new perspective on integrating topological data analysis with attention-based graph learning, opening new directions for robust and interpretable GNNs.

II. PRELIMINARIES

In this section, we briefly review the necessary background on Graph Neural Networks, attention mechanisms, and persistent homology. These concepts form the foundation of our proposed method.

A. Graph Neural Networks and Attention Mechanisms

We represent a graph as $G = (V, E)$, where V is the set of nodes with $|V| = n$, and $E \subseteq V \times V$ is the set of edges. Each node $v \in V$ is associated with a feature vector $x_v \in R^d$, and the feature matrix of all nodes is denoted by $X \in R^{n\times d}$. The graph's connectivity is encoded in an adjacency matrix $A \in R^{n*n}$, where $A_{uv} = 1$ if $(u, v) \in E$, and 0 otherwise. We denote the neighborhood of a node v by $N(v)$.

Graph Neural Networks (GNNs) operate on graph-structured data by iteratively updating node embeddings through message passing. At each layer l, a node aggregates information from its neighbors and updates its own representation:

$$\mathbf{h}_v^{(l+1)} = \sigma\left(\text{AGGREGATE}^{(l)}\left(\left\{\mathbf{h}_u^{(l)} : u \in \mathcal{N}(v) \cup \{v\}\right\}\right)\right),$$

where $\mathbf{h}_v^{(l)}$is the node embedding at layer l, $\text{AGGREGATE}^{(l)}$ is a permutation-invariant function (e.g., mean or sum), and σ is a non-linear activation function. This framework enables GNNs to incorporate both node features and structural information for various downstream tasks such as node classification and link prediction.

Graph Attention Networks (GATs) extend this framework by allowing nodes to assign different levels of importance to their neighbors using a learnable attention mechanism. Instead of uniformly aggregating neighbor features, GATs compute an attention score e_{uv} between a node v and each of its neighbors $u \in N(u)$ as follows:

$$e_{uv} = \text{LeakyReLU}\left(\mathbf{a}^\top[\mathbf{W}\mathbf{h}_u \,\|\, \mathbf{W}\mathbf{h}_v]\right),$$

where $\mathbf{W} \in \mathbb{R}^{d'\times d}$ is a learnable transformation matrix, $\mathbf{a} \in \mathbb{R}^{2d'}$ is a learnable attention vector, and $\|$ denotes vector concatenation. The attention scores are normalized across the neighborhood using a softmax function:

$$\alpha_{uv} = \frac{\exp(e_{uv})}{\sum_{k\in\mathcal{N}(v)} \exp(e_{kv})}.$$

The updated representation for node v is then given by:

$$\mathbf{h}_v^{(l+1)} = \sigma\left(\sum_{u\in\mathcal{N}(v)} \alpha_{uv}\mathbf{W}\mathbf{h}_u\right).$$

GATs can also employ multi-head attention to enhance learning stability and model expressiveness, where multiple attention mechanisms run in parallel and their outputs are concatenated or averaged.

Although GATs offer greater flexibility than traditional GNNs by learning to focus on the most relevant neighbors, they remain inherently limited to *local aggregation*. As a result, they may fail to detect global structural patterns such as symmetry, cycles, or higher-order regularities that are essential for understanding and detecting complex anomalies in graph data. This limitation motivates the integration of global topological features, as introduced in the next section.

B. Persistent Homology and Topological Features

Topological Data Analysis (TDA) provides a framework for understanding the global geometric and structural properties of data by studying its topological features such as connected components, holes, and higher-dimensional voids. Among the tools developed within TDA, *persistent homology* has emerged as a powerful method for summarizing topological features across multiple scales of resolution.

Given a graph $G = (V, E)$, we can construct a sequence of nested simplicial complexes called a *filtration* that captures how the topological structure evolves as we progressively include edges (or higher-order simplices) based on a filtration parameter, such as node degree, distances, or feature similarities. As the filtration progresses, new topological features may appear (e.g., a new connected component or cycle) and later disappear (e.g., when a cycle is filled in). Persistent homology tracks the birth and death of these features, producing a concise summary known as a *persistence diagram* or *barcode*.

Formally, persistent homology computes a multiscale summary of the k-dimensional homology groups H_k for $k = 0, 1, 2, \ldots$, where H_0 represents connected components, H_1 captures loops or cycles, and H_2 encodes voids. The output is a multiset of intervals $\{(b_i, d_i)\}$, where each pair corresponds to a topological feature that is born at filtration value b_i and dies at d_i. The difference $d_i - b_i$ represents the *persistence* of that feature, which intuitively measures its significance.

While classical TDA methods are not tailored specifically for graph-structured data, recent developments in *Graph Filtration Learning (GFL)* have emerged to bridge this gap by integrating topological reasoning directly into the graph learning process. Traditional persistent homology techniques apply fixed filtration strategies such as constructing Vietoris–Rips or clique complexes based on static distance metrics or edge weights without adapting to the task or learning signals. This rigidity limits their effectiveness when applied to diverse graph-based tasks like anomaly detection.

GFL addresses this limitation by introducing learnable filtration functions, which are parameterized by neural network modules and jointly optimized with downstream objectives. These functions assign real-valued filtration scores to graph elements (e.g., nodes or edges) based on their features or structural roles, effectively reordering the inclusion of simplices in the filtration sequence. By doing so, the filtration becomes data and task dependent, allowing the model to emphasize topologically relevant regions of the graph that are critical for a given learning objective.

During training, the model constructs multiple filtered simplicial complexes corresponding to different learned filtrations, from which persistent homology is computed. The resulting persistence diagrams capture multi-scale topological signatures, encoding information about the birth and death of connected components, cycles, and other high-order structures. These diagrams are then embedded into a vector space using differentiable representations such as persistence images, landscapes, or statistical descriptors, making them compatible with gradient-based optimization.

The integration of these vectorized topological features into graph neural networks provides a powerful enhancement: it enables the model to be both locally attentive (via message passing) and globally aware (via persistent topology). This dual capability improves the expressiveness, interpretability, and robustness of graph-based learning models, especially in domains where anomalies manifest as subtle structural variations rather than simple attribute outliers.

In our work, we extract persistent topological features from node neighborhoods and inject them into the GAT attention mechanism. This allows the model to become sensitive to not only local attribute-based patterns but also the global structural context in which a node is embedded enabling more effective detection of anomalies that manifest as topological irregularities.

III. RELATED WORK

Recent years have seen significant progress in graph-based machine learning, particularly in tasks involving node classification, link prediction, and anomaly detection. Graph Neural Networks (GNNs) have emerged as a leading approach for learning from relational data due to their ability to integrate node features with structural information. However, their application to anomaly detection still faces key limitations, especially when dealing with globally structured anomalies or adversarial manipulations. In this section, we review relevant prior work in three main areas: (1) graph-based anomaly detection methods, (2) Graph Attention Networks (GATs) and their use in security contexts, and (3) the integration of Topological Data Analysis (TDA), specifically persistent homology, into graph learning frameworks.

A. Graph-based Anomaly Detection

Anomaly detection on graphs plays a critical role in various real-world applications, including social behavior analysis, financial fraud detection, intrusion monitoring,

communication networks, and academic collaboration networks [1]. Due to the rich structural information encoded in graphs, anomaly detection in this domain poses unique challenges, particularly due to the non-Euclidean nature of graph representations. This has led to the development of Graph Anomaly Detection (GAD) techniques, which aims to identify anomalous components, such as nodes, edges, or subgraphs within a single graph, as well as detect anomalous graphs within a larger collection or database [2]. In this work, we specifically focus on the task of detecting node anomalies in attributed graphs, where both node features and structural information are leveraged to uncover irregular or suspicious behaviors.

Early approaches to GAD often relied on node-level heuristics (e.g., degree centrality) , statistical thresholds, or community-based measures [3]. However, these methods typically lack robustness and generalizability across diverse and noisy graph structures. More recent methods leverage Graph Neural Networks (GNNs) for anomaly detection, exploiting their ability to integrate node attributes with structural context [4]. Techniques include autoencoder-based models for node reconstruction [5], contrastive learning [6], [7], and outlier ranking based on embedding distances [8]. However, most of these approaches rely heavily on local neighborhood information and often fail to capture global structural irregularities, such as cycles, symmetries, or redundant motifs, that are essential for detecting coordinated or adversarial attacks.

B. Graph Attention Networks for Anomaly Detection

A considerable number of GNN-based approaches are categorized under network architectures that adopt either GCN or GAT as their foundational models. While GCN-based frameworks have shown strong performance in detecting anomalous nodes, their simple aggregation mechanisms often lead to the over-smoothing problem, which limits their ability to extract discriminative features. To address this limitation, GAT-based approaches have been proposed as an alternative, offering adaptive, attention-driven aggregation that better captures the relative importance of neighboring nodes.

Graph Attention Networks (GATs) extend traditional GNNs by enabling nodes to attend selectively to their neighbors via learnable attention coefficients. This improves robustness to noise and allows the model to prioritize informative connections. GATs have been employed in various anomaly detection tasks, including network traffic [9], IoT threat detection [10], and financial fraud detection [11]. These methods benefit from adaptive weighting but are still limited to local message passing, making them less effective in identifying anomalies that are global or distributed across the graph. Moreover, standard GATs do not explicitly consider higher-order structural patterns, which can be critical for detecting adversarial manipulations or policy violations.

C. Topological Data Analysis in Graph Neural Networks

Topological Data Analysis (TDA) is a powerful framework for uncovering geometric and topological structure in complex datasets, offering valuable insights that go beyond traditional statistical or machine learning techniques. Unlike these conventional approaches, TDA focuses on understanding the shape and connectivity of data as it lies on, or approximates, a lower-dimensional manifold embedded in a high-dimensional feature space [12]. Graphs, by nature, are topological objects that encode structural relationships through nodes and edges, making the integration of TDA into graph-based learning a promising direction for enhanced data representation [13], [14].

Enabling topology-aware capabilities within Graph Neural Networks (GNNs) is particularly beneficial for enriching node and graph representations with higher-order structural information. This integration can improve the expressiveness of GNNs and enhance performance across multiple tasks, including anomaly detection [15]. Recent studies have investigated methods to incorporate topological features into GNN-based architectures by leveraging persistent homology, a core tool in TDA, to capture multiscale structural patterns and preserve global graph properties [16]. By embedding such high-order topological information, these approaches aim to make GNNs more robust and discriminative, particularly in identifying complex or distributed anomalies.

D. Our Contribution

In contrast to prior work, we propose a novel architecture that integrates persistent topological features directly into the attention computation of Graph Attention Networks. This enables the model to leverage both local attribute-based signals and global structural information during message passing. By making GATs topologically aware, our method improves robustness, interpretability, and effectiveness in detecting anomalies, especially those rooted in global graph geometry. To the best of our knowledge, this is the first approach that jointly utilizes persistent homology and attention mechanisms for graph-based anomaly detection.

IV. METHODOLOGY

In this work, we propose a novel approach for anomaly detection on graphs by enhancing the Graph Attention Network (GAT) architecture with topological awareness. Our key idea is to incorporate topological features derived from persistent homology into the attention mechanism of GATs. This integration enables the model to consider not only local node features and structural connections but also global topological patterns, such as cycles and higher-order connectivity, during message passing. The resulting topology-aware GAT improves sensitivity to structural irregularities and enhances the model's ability to detect subtle or distributed anomalies.

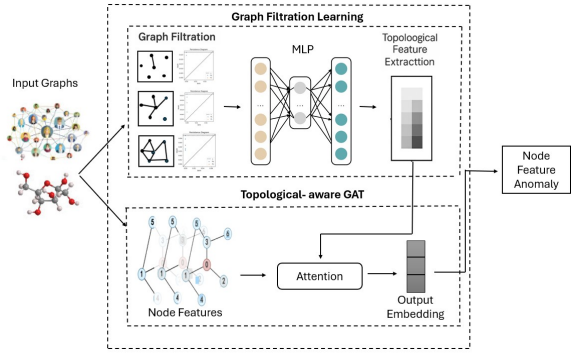

Fig. 1: General architecture of TopoGAT-based techniques on how the TDA is applied to leverage the performance of GAT

A. Problem Definition

Let $G = (V, E)$ be an attributed graph, where V is the set of nodes, $E \subseteq V \times V$ is the set of edges, and each node $v \in V$ is associated with a feature vector $\mathbf{x}_v \in \mathbb{R}^d$. The goal is to detect anomalous nodes in G based on both attribute and structural information. Formally, we aim to learn an anomaly score function $f : V \to \mathbb{R}$ that assigns a score to each node, reflecting its likelihood of being anomalous.

B. Overview of the Proposed Framework

Our proposed framework consists of three main components:

1) A topological feature extraction module that computes persistent homology on the graph to capture multiscale topological patterns.
2) A modified attention mechanism that incorporates both feature similarity and topological descriptors into the computation of attention coefficients.
3) A topology-aware GAT architecture that uses the enhanced attention scores for node representation learning and anomaly scoring.

C. Topological Feature Extraction

We first extract topological features from the graph using *persistent homology*, a fundamental tool from Topological Data Analysis (TDA). Given a graph G, we construct a filtration, such as Vietoris–Rips complex based on node distances. Persistent homology is then applied to track the birth and death of topological features (e.g., connected components and cycles) across this filtration. The resulting *persistence diagram* is vectorized using techniques such as persistence images or summary statistics (e.g., max persistence, average birth/death), yielding a topological feature vector $\mathbf{t}_v \in \mathbb{R}^k$ for each node v.

D. Topology-Aware Attention Mechanism

We modify the standard GAT attention mechanism to incorporate topological information. In the original GAT, the unnormalized attention score between a target node v and its neighbor $u \in \mathcal{N}(v)$ is computed as:

$$e_{uv} = \text{LeakyReLU}\left(\mathbf{a}^\top [\mathbf{W}\mathbf{h}_u \,\|\, \mathbf{W}\mathbf{h}_v]\right),$$

where $\mathbf{W}$ is a learnable linear transformation and $\mathbf{a}$ is a learnable attention vector. We enhance this score by integrating topological information as follows:

$$e_{uv}^{\text{topo}} = \text{LeakyReLU}\left(\mathbf{a}^\top [\mathbf{W}\mathbf{h}_u \,\|\, \mathbf{W}\mathbf{h}_v \,\|\, \mathbf{T}(\mathbf{t}_u, \mathbf{t}_v)]\right),$$

where $\mathbf{T}(\mathbf{t}_u, \mathbf{t}_v)$ is a topological descriptor function (e.g., concatenation) combining the persistence-based features of nodes u and v.

The attention coefficients are then normalized via softmax:

$$\alpha_{uv} = \frac{\exp(e_{uv}^{\text{topo}})}{\sum_{k \in \mathcal{N}(v)} \exp(e_{kv}^{\text{topo}})},$$

and the final node representation is computed as:

$$\mathbf{h}'_v = \sigma\left(\sum_{u \in \mathcal{N}(v)} \alpha_{uv} \mathbf{W}\mathbf{h}_u\right).$$

E. Anomaly Scoring

After obtaining the final node embeddings from the topology-aware GAT, we compute an anomaly score for each node based on a learned anomaly detection head (e.g., an MLP). Nodes with higher scores are considered to be more likely to be anomalous. In the supervised setting, we assume access to a labeled set of nodes for training:

$$\mathcal{D} = \{(v_i, y_i)\}_{i=1}^N, \quad y_i \in \{0, 1\},$$

where $y_i = 1$ indicates that node v_i is anomalous, and $y_i = 0$ denotes a normal node. After computing topology-aware node embeddings $\mathbf{h}'_v$ using our modified attention mechanism, we train a binary classifier to estimate the probability that a node is anomalous. This is implemented as a fully connected neural layer (e.g., MLP) followed by a sigmoid activation:

$$f(v) = \text{MLP}(\mathbf{h}'_v) \in [0, 1],$$

where $f(v)$ represents the anomaly score for node v. Higher scores indicate a higher likelihood of being anomalous.

The model is trained using the binary cross-entropy loss:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N \left[y_i \log(f(v_i)) + (1 - y_i) \log(1 - f(v_i))\right].$$

To address class imbalance, common in anomaly detection, we apply a weighted version of the loss function:

$$\mathcal{L}_{\text{w}} = -\sum_{i=1}^N \left[w_1 y_i \log(f(v_i)) + w_0 (1 - y_i) \log(1 - f(v_i))\right],$$

where w_1 and w_0 are class-specific weights, typically chosen inversely proportional to class frequencies.

During inference, the trained model outputs $f(v)$ as a soft anomaly score for each node. A threshold τ can be selected (e.g., based on validation set performance) to convert the score into a binary decision:

$$\text{Label}(v) = \begin{cases} 1 & \text{if } f(v) \geq \tau \\ 0 & \text{otherwise} \end{cases}.$$

This supervised approach enables the model to learn anomaly-relevant patterns in both node attributes and graph topology through the end-to-end training of the topology-aware attention mechanism.

F. Model Summary

By enriching the attention mechanism with persistent topological features, our approach enables GATs to attend to structurally significant regions of the graph that may otherwise be indistinguishable using local feature similarity alone. This design improves the model's robustness to noise, enhances interpretability, and yields superior performance in detecting subtle or globally structured anomalies.

V. EXPERIMENT

The goal of our experiments is to evaluate the effectiveness of incorporating topological features derived via Graph Filtration Learning (GFL) into the attention mechanism of Graph Attention Networks (GAT) for anomaly detection in graphs. Specifically, we aim to answer the following research questions:

- How does topology-aware attention compare to standard GAT in capturing structural irregularities?
- Can persistent topological features improve the performance of GATs in detecting anomalous nodes?

A. Datasets

We conduct experiments on the following datasets:

- **Synthetic Graphs:** We generate random graphs using the Erdős–Rényi model, where node features are assigned randomly. Anomalous substructures are embedded explicitly to serve as ground truth for evaluating anomaly detection.
- **Cora:** A citation network comprising 2,708 nodes, 5,429 edges, and 1,433-dimensional binary bag-of-words features. Anomalies are synthetically injected by perturbing node features and randomly rewiring edges.
- **Amazon Computers:** A real-world co-purchase network containing approximately 13,000 nodes and 245,000 undirected edges. Each node is associated with a 767-dimensional bag-of-words feature vector. We simulate anomalies by corrupting node features to assess the model's ability to detect structural and semantic outliers.

Each dataset is partitioned into training, validation, and test subsets with known anomaly labels. We train the model using a binary cross-entropy loss to classify nodes as normal or anomalous.

B. Baselines

We compare our proposed method against the following baselines:

- **GCN** [4]: Standard graph convolutional network.
- **GAT** [17]: Graph attention network without topological enhancements.
- **TOGL** [16]: Uses persistent homology and MLPs without GNNs.
- **TopoGAT**: Our topology-aware attention model combining GFL and MLP-enhanced attention.

C. Evaluation Metrics

We evaluate performance using the following metrics:

- **AUC-ROC:** Area under the receiver operating characteristic curve.
- **AUC-PR:** Area under the precision-recall curve.
- **F1-Score:** Harmonic mean of precision and recall at optimal threshold.

D. Implementation Details

We compute persistent homology using the `ripser.py` library. For each node, we construct local subgraphs based on graph filtration thresholds derived from centrality metrics. Specifically, we use three node level measures *degree*, *closeness*, and *betweenness centrality* and define threshold levels at the 25th, 50th, and 75th percentiles of each distribution. For each threshold, a subgraph is extracted by including the node and its neighbors whose metric values exceed the threshold.

We compute the pairwise distance matrix based on 2D graph layouts (using `spring_layout`) and apply Vietoris-Rips filtration on each subgraph to obtain persistence diagrams. These diagrams are vectorized into fixed-length representations by truncating or padding the top 10 most persistent 1-dimensional features (loops), resulting in a 20-dimensional vector per filtration. The final topological feature vector for each node concatenates all these vectors across all filtration metrics and thresholds.

These topological features are passed through an MLP encoder with two ReLU-activated layers of size 128 and 32, respectively. The output is concatenated with original node features and fed into a 2-layer GAT with 8 attention heads and 8 hidden units per head. Models are trained for 100 epochs using the Adam optimizer with a learning rate of 0.0005. All experiments are repeated five times with different random seeds, and we report the average and standard deviation of AUC, F1, and AUC-PR scores.

E. Results and Analysis

Discussion of Attention Coefficients with and without Topological Modulation: To understand the impact of topology-aware attention, we compared the distribution of attention coefficients in two settings: (i) a baseline Graph Attention Network (GAT) that relies solely on node feature

Method/Dataset	SYNTHETIC GRAPH			CORA			AMAZON		
	AUC-ROC	F1-Score	AUC-PR	AUC-ROC	F1-Score	AUC-PR	AUC-ROC	F1-Score	AUC-PR
GCN	0.468	0.01	0.09	0.406	0.19	0.08	0.291	0.03	0.04
GAT	0.627	0.2	0.18	0.532	0.19	0.18	0.451	0.05	0.08
TOGL	0.757	0.25	0.2	0.613	0.22	0.19	0.534	0.08	0.11
TopoGAT	**0.872**	**0.3**	**0.25**	**0.742**	**0.38**	**0.30**	**0.670**	**0.12**	**0.15**

TABLE I: Performance comparison (AUC-ROC, F1 Score, and AUC-PR) of anomaly detection methods across three datasets.

representations, and (ii) our proposed TopoGAT model, which incorporates MLP-encoded topological features derived from multiple graph filtrations (degree, closeness, and betweenness centrality). As shown in Figure 2, we visualize the normalized degree, closeness, and betweenness centrality values used for filtration.

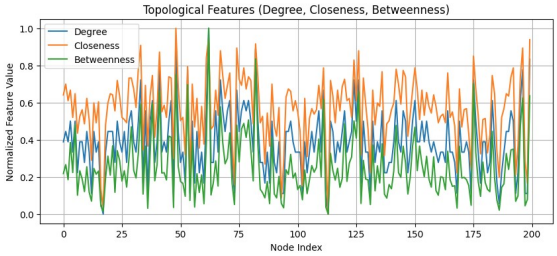

Fig. 2: Normalized topological features computed for each node in the synthetic graph. The features correspond to node degree, closeness centrality, and betweenness centrality, which are used as filtration functions in our topological augmentation.

In Figure 3 the baseline GAT yields a relatively narrow and uniform distribution of attention scores, indicating that it treats most neighbors with comparable importance. In contrast, the attention coefficients in TopoGAT exhibit a more diverse and skewed distribution, reflecting stronger differentiation among nodes based on their structural roles in the graph. This suggests that the addition of topological priors enables the model to focus attention on salient or anomalous nodes more effectively. Such modulation is particularly beneficial for anomaly detection tasks, where structural irregularities may not be apparent in raw node features alone.

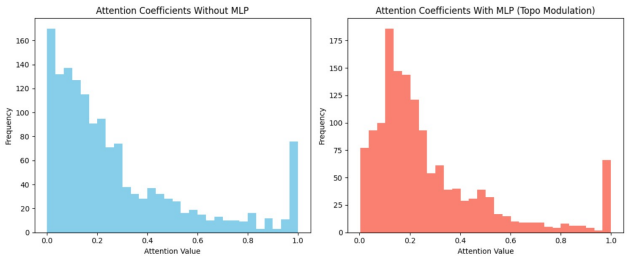

Fig. 3: Histogram of attention coefficients on synthetic graph data. The left panel shows a plain GAT trained on raw features, while the right shows TopoGAT with MLP-encoded topological features.

Analysis of the Performance of TopoGAT Model in Detecting Anomalous Nodes: This section evaluates the effectiveness of Graph Attention Networks (GATs) in identifying anomalous nodes within a graph structure. By comparing models with and without topological feature integration, we assess how attention mechanisms contribute to the detection of structural irregularities.

As shown in Table 1, the TopoGAT model demonstrates consistently strong performance across all three datasets (Synthetic Graph, CORA, and Amazon) outperforming baseline models in nearly every metric. On the synthetic dataset, TopoGAT achieves the highest AUC-ROC (0.872), F1-score (0.30), and AUC-PR (0.25), clearly surpassing GCN, GAT, and TOGL. This indicates TopoGAT's effectiveness in capturing topological irregularities and distinguishing anomalous patterns in graph-structured data.

On the CORA dataset, TopoGAT also performs best, with an AUC-ROC of 0.742, an F1-score of 0.38, and an AUC-PR of 0.30. These results show a significant margin over TOGL, which achieves moderate scores, and GCN/GAT, which plateau around 0.19 F1. The improved AUC-PR further highlights TopoGAT's ability to handle class imbalance and detect subtle anomalies in citation networks.

For the more challenging Amazon dataset, TopoGAT still leads with an AUC-ROC of 0.670 and an F1-score of 0.12, outperforming the next best model, TOGL. While the AUC-PR remains relatively low (0.15), it is notably higher than all other methods, including GAT and GCN. This suggests that topological augmentation benefits even large-scale, real-world datasets with sparse labels and subtle anomalies.

Overall, TopoGAT's superior results across diverse settings confirm the advantage of integrating persistent topological features with graph attention mechanisms for graph-based anomaly detection.

VI. CONCLUSION

In this work, we proposed a topology-aware anomaly detection framework that integrates node-level topological features, derived through filtration and persistent homology, into a Graph Attention Network. By leveraging graph-based metrics such as degree, closeness, and betweenness centrality, we constructed filtration functions that capture the structural roles of nodes. Persistence diagrams were extracted and encoded to enrich node representations with higher-order topological information.

Our experimental results across multiple datasets, including synthetic graphs, Cora, and Amazon, demonstrate that the proposed TopoGAT model consistently outperforms baseline GCN and GAT models in terms of AUC, F1, and AUC-PR scores. These findings highlight the value of incorporating topological signals into graph learning architectures for more robust anomaly detection.

Future work will explore the integration of more complex topological invariants and apply the approach to dynamic and heterogeneous graphs.

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